Statistical Games: Computing Minimax Rules

STAT 220B – Lecture 8

Where we left off

Lectures 6 and 7 built the game-theoretic foundation:

- Two-person zero-sum games, randomized strategies, upper and lower values.
- Verification techniques (Theorems 1–4) via guessed priors and equalizer strategies.
- The Minimax Theorem (Theorem 15): for finite Θ and bounded loss, a value exists and minimax strategies are guaranteed.

Today we apply this machinery to **statistical decision problems**, where Player I is nature choosing $\theta \in \Theta$ and Player II is the statistician choosing a decision rule $\delta(X)$ based on data $X \sim P_\theta$.

The Minimax Theorem says minimax rules exist. The remaining question is **how to compute them**.

Reading: Berger, Section 5.3 (especially 5.3.1 and 5.3.2).

Statistical games as game theory (Berger 5.3.1)

The translation

A statistical decision problem becomes a two-person zero-sum game by reading the components into the game-theoretic framework:

Game-theoretic notion	Statistical interpretation
Player I's strategy θ	True parameter value
Player II's strategy δ^*	Randomized decision rule
Loss $L(\theta, \delta^*)$	Frequentist risk $R(\theta, \delta^*) = \mathbb{E}_{P_\theta}[L(\theta, \delta^*(X))]$
Player I's randomization π	Prior distribution on Θ
$L(\pi, \delta^*) = \mathbb{E}^\pi[L(\theta, \delta^*)]$	Bayes risk $r(\pi, \delta^*) = \mathbb{E}^\pi[R(\theta, \delta^*)]$
Maximin strategy π^M	**Least favorable prior**
Minimax strategy δ^{*M}	Minimax decision rule

The terminology changes (we say "least favorable prior" instead of "maximin strategy") because nature is not really an intelligent opponent. The mathematics is identical.

What makes statistical games hard

In a finite matrix game, the risk set $S \subseteq \mathbb{R}^m$ is a convex polytope, easy to draw and search.

In a statistical game, several features make things harder:

- Θ may be continuous. The risk function $R(\cdot, \delta^*) : \Theta \to \mathbb{R}$ lives in an infinite-dimensional function space.
- **The decision rule space** $\mathcal{D}^*$ is enormous. Even for $X \sim \mathcal{N}(\theta, 1)$, $\mathcal{D}^*$ contains all measurable functions $\mathbb{R} \to \mathcal{A}$, plus randomizations.
- **The risk set** S is not directly accessible. Plotting it like a polytope is rarely an option.

So the geometric S-game picture is conceptually correct but not computationally useful in this setting. We need different tools.

Three methods (Berger $§$5.3.2)

Berger lays out three approaches for finding minimax rules in statistical games.

Method 1: Direct calculation. Reduce $\mathscr{D}^*$ to a small class (via admissibility, sufficiency, or invariance considerations), then compute $\sup_\theta R(\theta, \delta^*)$ for each remaining rule and minimize. Works when the reduction makes $\mathscr{D}^*$ tractable. Standard in testing problems where UMP tests reduce the search.

Method 2: Guess a least favorable prior. Find π^M, compute the Bayes rule δ^{π^M}, verify it is minimax via a sufficient condition (Theorem 17 or 18).

Method 3: Guess an equalizer rule. Search for a rule with constant risk $R(\theta, \delta) \equiv C$. If such a rule is also Bayes, Theorem 17 gives minimaxity for free.

Today we develop methods 2 and 3. Method 1 will surface in Chapter 8 (testing) and Chapter 6 (invariance).

The equalizer-Bayes theorem (Theorem 17)

Statement

Theorem 17 (Berger $§$5.3.2)

Suppose δ^π is Bayes with respect to a prior π, and

$$R(\theta, \delta^\pi) \;\le\; r(\pi, \delta^\pi) \qquad \text{for all } \theta \in \Theta. \qquad (*)$$

Then δ^π is **minimax**, and π is a **least favorable prior**.

Condition $(*)$ says: the worst-case frequentist risk of δ^π is no larger than its average risk under π. Two ways this can happen:

- **Equalizer rule.** If $R(\theta, \delta^\pi)$ is constant in θ, then $R(\theta, \delta^\pi) = r(\pi, \delta^\pi)$ automatically, and $(*)$ holds with equality. This is the cleanest case.
- **Concentrated worst case.** If π puts all its mass on the points θ where $R(\theta, \delta^\pi)$ attains its supremum, the inequality holds with equality at the support of π.

Either way, the prior is concentrated on the "hard" regions of Θ, which is what makes it least favorable.

Proof

Let δ be any decision rule. Three observations:

(a) Worst-case risk dominates average risk:

$$\sup_\theta R(\theta,\delta) \;\ge\; \int_\Theta R(\theta,\delta)\,d\pi(\theta) \;=\; r(\pi,\delta).$$

(b) δ^π minimizes the Bayes risk against π:

$$r(\pi,\delta) \;\ge\; r(\pi,\delta^\pi).$$

(c) By the hypothesis $(*)$:

$$r(\pi,\delta^\pi) \;\ge\; \sup_\theta R(\theta,\delta^\pi).$$

Chaining (a), (b), (c):

$$\sup_\theta R(\theta,\delta) \;\ge\; \sup_\theta R(\theta,\delta^\pi).$$

Since δ was arbitrary, δ^π achieves the infimum of $\sup_\theta R(\theta,\delta)$ over all rules. So δ^π is minimax. $\square$

The statement about π being least favorable uses the same chain, this time supping over π' on the maximin side; we omit the details.

Corollary: constant-risk Bayes rules

Corollary

If δ^π is a Bayes rule with respect to π and has **constant risk** $R(\theta, \delta^\pi) = C$ for all θ, then δ^π is minimax.

Proof. Constant risk C gives $r(\pi, \delta^\pi) = \int C\, d\pi = C$, so $R(\theta, \delta^\pi) = C = r(\pi, \delta^\pi)$ for all θ. Apply Theorem 17. $\square$

The practical recipe.

1. Fix a parametric family of priors π_α (e.g., conjugate priors).
2. Compute the Bayes rule δ^{π_α} in closed form.
3. Compute $R(\theta, \delta^{\pi_\alpha})$ as a function of θ and α.
4. Solve for α that makes R constant in θ.
5. Conclude that δ^{π_α} is minimax.

This is the workhorse for "nice" statistical problems where conjugacy makes step 2 tractable.

Example 17: Binomial estimation (Berger $§$5.3.2)

Setup

Let $X \sim \text{Binomial}(n, p)$, with $p \in \Theta = [0, 1]$ unknown. Estimate p under squared-error loss $L(p, a) = (p - a)^2$.

Why this matters. Estimating a proportion is the most basic statistical problem: A/B test conversion rates, election polling, defect rates in manufacturing, vaccine efficacy. The MLE $\hat{p} = X/n$ is the default; we ask whether it can be improved in worst-case risk.

The plan. Apply Theorem 17's recipe with the conjugate $\text{Beta}(\alpha, \beta)$ prior. Find (α, β) making the resulting Bayes rule's risk constant in p.

Step 1: Bayes rule under Beta(α, β) prior

The Beta(α, β) prior is $\pi(p) \propto p^{\alpha-1}(1-p)^{\beta-1}$ on $[0, 1]$. With Binomial likelihood, the posterior is conjugate:

$$p \mid X = x \;\sim\; \text{Beta}(\alpha + x, \beta + n - x).$$

Under squared-error loss, the Bayes rule is the posterior mean. For Beta(α', β') the mean is $\alpha'/(\alpha' + \beta')$, so

$$\delta^\pi(x) \;=\; \frac{\alpha + x}{\alpha + \beta + n}.$$

Let $c = \alpha + \beta$. Then $\delta^\pi(x) = (\alpha + x)/(c + n)$ is linear in x.

Step 2: Risk function

Compute $R(p, \delta^\pi) = \mathbb{E}_p[(\delta^\pi(X) - p)^2]$.

$$\delta^\pi(X) - p \;=\; \frac{\alpha + X - p(c+n)}{c+n} \;=\; \frac{(X - np) + (\alpha - pc)}{c+n}.$$

The term $(X - np)$ is a mean-zero random variable; the term $(\alpha - pc)$ is a deterministic shift. Squaring and taking expectation:

$$R(p, \delta^\pi) \;=\; \frac{1}{(c+n)^2}\,\mathbb{E}_p\big[(X-np)^2 + 2(X-np)(\alpha - pc) + (\alpha - pc)^2\big].$$

The cross term has expectation zero. The remaining terms give:

$$R(p, \delta^\pi) \;=\; \frac{\mathrm{Var}_p(X) + (\alpha - pc)^2}{(c+n)^2} \;=\; \frac{np(1-p) + (\alpha - pc)^2}{(c+n)^2}.$$

Step 3: Solving for constant risk

Expand the numerator as a polynomial in p:

$$np(1-p) + (\alpha - pc)^2 \;=\; \alpha^2 + p\,(n - 2\alpha c) + p^2\,(c^2 - n).$$

For $R(p, \delta^\pi)$ to be constant in p, the coefficients of p and p^2 must both vanish:

$$c^2 - n = 0 \quad \text{and} \quad n - 2\alpha c = 0.$$

Solving:

$$c = \sqrt{n}, \qquad \alpha = \frac{n}{2c} = \frac{\sqrt{n}}{2}.$$

So $\alpha = \sqrt{n}/2$ and $\beta = c - \alpha = \sqrt{n}/2$. The constant-risk Bayes rule is

$$\boxed{\delta^{*M}(x) \;=\; \frac{x + \sqrt{n}/2}{n + \sqrt{n}}.}$$

Step 4: The minimax estimator and its risk

The constant risk value:

$$R(p, \delta^{*M}) \;=\; \frac{\alpha^2}{(c+n)^2} \;=\; \frac{n/4}{(\sqrt{n}+n)^2} \;=\; \frac{n/4}{n(1+\sqrt{n})^2} \;=\; \frac{1}{4(1+\sqrt{n})^2}.$$

By Theorem 17's corollary, δ^{*M} is **minimax**, and the Beta$(\sqrt{n}/2, \sqrt{n}/2)$ prior is **least favorable**.

Comparison with the MLE

The MLE is $\hat{p}(x) = x/n$, with risk $R(p, \hat{p}) = p(1-p)/n$. Worst case at $p = 1/2$: $R = 1/(4n)$.

Compare worst-case risks:

n	$\sup R(\hat{p})$	$\sup R(\delta^{*M})$	Ratio
1	0.2500	0.0625	4.00
10	0.0250	0.0144	1.73
100	0.00250	0.00207	1.21
1000	0.000250	0.000235	1.06

For small samples the minimax estimator has dramatically lower worst-case risk. For large n the two are nearly identical.

Interpretation. $\delta^{*M}(x) = (x + \sqrt{n}/2)/(n + \sqrt{n})$ adds $\sqrt{n}/2$ pseudo-successes and $\sqrt{n}/2$ pseudo-failures to the count, shrinking the estimate toward $1/2$. This is the formal justification for "add a half" style estimators in small-sample proportions.

When Theorem 17 fails: improper priors

The problem

For many natural statistical problems, the obvious candidate for a least favorable prior is **improper**: e.g.,

$$\pi(\theta) = 1 \quad \text{for } \theta \in \mathbb{R} \text{ (Lebesgue on the real line).}$$

If π is improper, $\int d\pi(\theta) = \infty$, so:

- The Bayes risk $r(\pi, \delta) = \int R(\theta, \delta)\, d\pi(\theta)$ is typically ∞ for every rule.
- δ^π is a **generalized** Bayes rule (minimizing posterior expected loss for each x).
- Theorem 17's hypothesis $R(\theta, \delta^\pi) \leq r(\pi, \delta^\pi) = \infty$ is vacuous.

We need a different mechanism. The standard fix is to approximate π by a sequence of proper priors and pass to the limit.

Theorem 18: sequences of priors

Theorem 18 (Berger $§§5.3.2)

Suppose $\{\pi_n\}$ is a sequence of proper priors with Bayes rules δ^{π_n} and finite Bayes risks $r(\pi_n) = r(\pi_n, \delta^{\pi_n})$. Let δ^* be a decision rule with

$$R(\theta, \delta^*) \leq \lim_{n\to\infty} r(\pi_n) < \infty \quad \text{for all } \theta \in \Theta. \qquad (**)$$

Then δ^* is minimax.

Proof. Let δ be any rule. For each n:

$$\sup_\theta R(\theta, \delta) \geq r(\pi_n, \delta) \geq r(\pi_n)$$

where the first inequality is "sup dominates average" and the second uses that δ^{π_n} minimizes Bayes risk against π_n. Taking $n \to \infty$:

$$\sup_\theta R(\theta, \delta) \geq \lim_n r(\pi_n) \geq \sup_\theta R(\theta, \delta^*),$$

the last step by hypothesis $(**)$. So δ^* is minimax. $\square$

How to use Theorem 18 in practice

A rule δ^* satisfying the bound in $(**)$ is called **extended Bayes**.

The recipe:

1. **Guess.** Identify the improper prior π suggested by the problem (often a noninformative prior from Berger Chapter 3).
2. **Choose a sequence.** Pick proper priors π_n such that some rescaling $c_n \pi_n \to \pi$ (e.g., $\mathcal{N}(0, n)$ converging to Lebesgue as $n \to \infty$).
3. **Compute** $r(\pi_n)$. Bayes risks of δ^{π_n} against π_n.
4. **Take the limit.** $\lim_n r(\pi_n)$, if it exists.
5. **Verify** $(**)$. Show that the natural candidate δ^* has risk bounded by this limit. Often δ^* has constant risk equal to the limit, in which case the inequality is an equality.

Example 15: Normal mean estimation (Berger $§$5.3.2)

Setup

Let $\bar{X} \sim \mathcal{N}(\theta, \sigma^2/n)$ with $\theta \in \Theta = \mathbb{R}$ unknown, σ^2 known. Estimate θ under squared-error loss $L(\theta, a) = (\theta - a)^2$.

Why this matters. This is the cornerstone estimation problem in classical statistics. The sample mean $\bar{X}$ is the MLE, the UMVUE, and (we will show) the minimax estimator.

The plan. Try the improper prior $\pi(\theta) = 1$. Theorem 17 will fail (Bayes risks are infinite), so use the sequence $\pi_n = \mathcal{N}(0, n)$, which approximates Lebesgue as $n \to \infty$. Verify the sample mean is extended Bayes via Theorem 18.

Step 1: Bayes rule under $\mathcal{N}(0, \tau^2)$ prior

For $\pi_\tau = \mathcal{N}(0, \tau^2)$:

Posterior precision: $\frac{n}{\sigma^2} + \frac{1}{\tau^2}$. Posterior variance: $V_\tau = \left(\frac{n}{\sigma^2} + \frac{1}{\tau^2}\right)^{-1} = \frac{\sigma^2 \tau^2}{n\tau^2 + \sigma^2}$.

Posterior mean (Bayes rule under squared-error loss):

$$\delta^{\pi_\tau}(\bar{x}) \;=\; \frac{n\tau^2}{n\tau^2 + \sigma^2}\,\bar{x}.$$

Let $c_\tau = n\tau^2/(n\tau^2 + \sigma^2)$. As $\tau \to \infty$, $c_\tau \to 1$, this goes to the sample mean.

Step 2: Bayes risk

For squared-error loss, the Bayes risk equals the expected posterior variance, which here is simply the posterior variance (constant in $\bar{x}$):

$$r(\pi_\tau) \;=\; r(\pi_\tau, \delta^{\pi_\tau}) \;=\; V_\tau \;=\; \frac{\sigma^2\tau^2}{n\tau^2 + \sigma^2}.$$

Dividing numerator and denominator by τ^2:

$$r(\pi_\tau) \;=\; \frac{\sigma^2}{n + \sigma^2/\tau^2}.$$

As $\tau \to \infty$:

$$\lim_{\tau\to\infty} r(\pi_\tau) \;=\; \frac{\sigma^2}{n}.$$

Step 3: Risk of the sample mean

The candidate $\delta^*(\bar{x}) = \bar{x}$ has risk

$$R(\theta, \delta^*) \;=\; \mathbb{E}_\theta[(\bar{X} - \theta)^2] \;=\; \mathrm{Var}_\theta(\bar{X}) \;=\; \frac{\sigma^2}{n} \quad \text{for all } \theta \in \mathbb{R}.$$

The risk is constant and equals the limiting Bayes risk:

$$R(\theta, \delta^*) \;=\; \frac{\sigma^2}{n} \;=\; \lim_{\tau \to \infty} r(\pi_\tau).$$

So the bound $(**)$ holds with equality for every θ.

Step 4: Apply Theorem 18

Choose the sequence $\pi_k = \mathcal{N}(0, \tau_k^2)$ with $\tau_k \to \infty$ (e.g., $\tau_k = k$). The Bayes risks satisfy

$$\lim_{k \to \infty} r(\pi_k) \;=\; \frac{\sigma^2}{n}.$$

The sample mean $\delta^*(\bar{x}) = \bar{x}$ has constant risk $R(\theta, \delta^*) = \sigma^2/n$, equal to the limit. So the bound $(**)$ holds with equality for all θ. By Theorem 18:

Conclusion

The sample mean $\bar{X}$ is minimax for estimating θ in $\bar{X} \sim \mathcal{N}(\theta, \sigma^2/n)$ under squared-error loss.

The "least favorable prior" is the improper $\pi(\theta) = 1$ on $\mathbb{R}$, in the extended-Bayes sense: the noninformative prior is the limit of priors that progressively wash out information about θ.

Summary

What we built today

Statistical games ($§$5.3.1). The decision problem is a two-person zero-sum game with Θ, $\mathcal{D}^*$, and risk $R(\theta, \delta^*)$. The maximin strategy is the least favorable prior. The Minimax Theorem guarantees existence; the task today was computation.

Theorem 17. If δ^π is Bayes with respect to π and $R(\theta, \delta^\pi) \leq r(\pi, \delta^\pi)$ for all θ, then δ^π is minimax. The corollary: a constant-risk Bayes rule is minimax.

Example 17 (Binomial). With $X \sim \text{Binomial}(n, p)$ and squared loss, the Beta$(\sqrt{n}/2, \sqrt{n}/2)$ prior produces the constant-risk Bayes rule

$$\delta^{*M}(x) = \frac{x + \sqrt{n}/2}{n + \sqrt{n}},$$

which is minimax. Dramatically better than the MLE for small n.

Theorem 18. Sequence version. For improper priors, approximate by proper π_n and require $R(\theta, \delta^*) \leq \lim_n r(\pi_n)$ for all θ.

Example 15 (Normal mean). $\bar{X} \sim \mathcal{N}(\theta, \sigma^2/n)$ with squared loss: the sample mean has constant risk σ^2/n equal to $\lim_{\tau \to \infty} r(\mathcal{N}(0, \tau^2))$. The sample mean is minimax.

Reading

Berger, Section 5.3 (sections 5.3.1 and 5.3.2). Examples 14, 15, 17 worked in detail; Example 16 also worth reading for the testing version of Theorem 17.